PRODUCT REQUIREMENTS DOCUMENT (PRD)

Nutro — Intelligent Food & Fluid Architecture

Domain: nutro-ai.app **Status:** Phase 1 Planning & Design Baseline **Author:** Product Orchestrator **Target Launch:** iOS / Android Native

1. Executive Summary & Core Problem Space

Nutrition and hydration applications suffer from severe user fatigue. Competitors either present overwhelming manual input spreadsheets or narrow single-metric silos. Cal AI proved the viral demand for visual meal recognition, while Water Llama validated the power of character-driven gamification. **Nutro (`nutro-ai.app`)** combines these dual modalities into a singular unified ecosystem: combining frictionless visual AI logging with tactile liquid physics, pantry-aware kitchen suggestions, and family synchronization.

THE USER FAILURE LOOP

- Manual database lookups take 60+ seconds per meal.
- Water tracking feels like a chore without real-time emotional feedback.
- Fridge food waste: users do not know what to cook with available leftovers.
- Families lack visibility over children's and elder hydration habits.

THE NUTRO INTERVENTION

- Sub-2-second vision food logs with immediate macro parsing.
- Tactile liquid character and haptic fill-physics inspired by Water Llama.
- "Fridge AI": Snap fridge/pantry to unlock macro-balanced local recipes.
- Nutro Home: Orbit-style multi-profile management for households.

2. Market Positioning & Differentiator Matrix

Feature Dimension	Cal AI Baseline	Water Llama Baseline	Nutro (nutro-ai.app)
Primary Focus	Food & calorie scanning only	Fluid logging only	**Unified Macro & Hydration Ecosystem**
Visual Engine	Minimal monochrome cards	Illustrated cartoon animals	**Sleek B&W Core + Dynamic Fluid & Macro Accents**
Culinary Intelligence	Passive past logging	None	**"What Can I Eat?" Fridge Vision + Recipe Importer**
Social & Family	Isolated single user	Friend streaks only	**Nutro Home ($5) & Home+ ($10) Admin Orbit**
Regional Diet Adaptability	Primarily Western foods	Metric/Imperial volume	**Localized cuisine detection (e.g., Desi, Asian, Med)**

3. Detailed 5 Core Features & Secondary Capabilities

Nutro's product architecture is split into five daily-driver utilities supported by social and family extensions designed to drive organic daily active retention (DAR).

Module	Functional Specification	User Value & Edge Case Handling
1. Food AI Tracking	Camera scan or voice note $\rightarrow$ Supabase storage $\rightarrow$ Gemini/OpenRouter vision. Returns calories, protein, carbs, fat.	Low-confidence detector (<85%) triggers one-tap manual stepper sheet. Eliminates typing.
2. Tactile Fluid Tracker	Interactive cup/character container with fluid wave shaders via React Native Skia & Reanimated.	Multi-beverage options (water, tea, coffee, electrolytes). Haptic detents on every 100ml interval.
3. "What Can I Eat?"	Pantry/fridge image recognition. Filters dish generation strictly against user's remaining daily calories.	Prevents bingeing and stops food waste by generating 3 step-by-step cooking options with pantry matches.
4. Predictive Calendar	Dual-track daily matrix. Visualizes caloric balance against hydration curve. Highlights deficit streaks.	Identifies fatigue patterns (e.g., "Protein consistently down on Thursdays") with actionable nudges.
5. Universal Recipe Hub	Parses YouTube URLs, web links, or handwritten recipe photos into structured step cards with timers.	Transforms passive entertainment into actionable cooking workflows with verified nutrition badges.
Secondary Extensions	• Nutro Home ($5) & Home+ ($10) • Smart Grocery Checklist • Lock Screen & Home Screen Widgets	• Admin orbit ring tracks family wellness. • Auto-populates missing recipe ingredients. • 1-tap quick water log from iOS lock screen.

4. Visual Design Language & Color System

The interface implements a **Minimal High-Contrast Dark Foundation** (Cal AI inspired) paired with **Functional Fluid Color Accents** (Water Llama inspired). Color is never decorative; it communicates state.

MONOCHROME CORE SURFACE

- **Background:** OLED Deep Black (#000000)
- **Card Surface:** Graphite Grey (#121214)
- **Borders & Outlines:** Deep Zinc (#242428)
- **Primary Action CTA:** Solid White Button with Black Typography for peak contrast.

FUNCTIONAL ACCENT SPECTRUM

- **Aqua Cyan (#38BDF8):** Water logging, hydration levels, droplet animations.
- **Energy Lime (#A3E635):** Caloric progress and deficit indicator.
- **Tangerine (#FB923C):** Kitchen recipe cards & food badges.
- **Royal Indigo (#A855F7):** Nutro Home family orbit indicators.

5. The 20-Screen High-Conversion Onboarding Blueprint

Designed around the single-focus philosophy: one question per viewport with large tactile selectors, haptic wheels, and zero cognitive clutter.

Stage	Screen Title & Topic	Interactive UI Pattern	Data Field Bound

Hook	01. Splash & Value Hook
	02. Primary Objective
	03. Target Pace
		• Mascot Hello + Soundbite
		• 4 Macro Cards (Strength, Cut, Health)
		• Pace Slider (Steady vs. Fast)
			`auth_intent`
			`goal_type`
			`target_speed_kg_wk`

Biometrics	04. Biological Sex
	05. Date of Birth
	06. Height Selector
	07. Current Weight
	08. Target Weight
	09. Activity Profile
		• Binary + Non-Binary Tiles
		• iOS Haptic Drum Picker
		• Vertical Sliding Ruler (cm / ft)
		• Horizontal Stepper Gauge (kg / lbs)
		• Calibrated Target Dial
		• 4 Segment Cards (Sedentary to Heavy)
			`sex`
			`dob`
			`height_cm`
			`current_weight_kg`
			`target_weight_kg`
			`activity_level`

Context & Diet	10. Location / Climate
	11. Dietary Restrictions
	12. Regional Cuisines
	13. Meal Frequency
	14. Hydration Baseline
		• Auto-Geo or City Search (Ahmedabad, etc.)
		• Multi-tag pills (Veg, Jain, Vegan, Keto)
		• Regional cuisine tiles (Gujarati, Med, etc.)
		• Stepper (2 to 5 meals/day)
		• Quick glass count picker
			`city, climate_factor`
			`diet_style`
			`regional_cuisines[]`
			`meals_per_day`
			`current_water_ml`

Micro-Demo & Paywall	15. "Aha!" Micro-Trial
	16. Frictionless Auth Gate
	17. Health & Notifications
	18. Plan Generation Engine
	19. Personalized Target Reveal
	20. Value Matrix & Paywall
		• Interactive Liquid Cup (Swipe to fill 250ml)
		• Sign in with Apple / Google (Forced Auth)
		• Permission toggles + Widget install hook
		• Algorithmic spinner (BMR calculation)
		• Calorie/Water target curve display
		• Home vs. Home+ pricing with trial toggle & [X]
			`session_trial_completed`
			`user_id (Supabase)`
			`push_token, sync_health`
			`bmr_calculated`
			`daily_targets`
			`subscription_tier`

6. Technical Implementation Topology

• **Mobile Frontend:** React Native with Expo Router and NativeWind v4 (Tailwind CSS for native runtime). Fast development directly inside Antigravity IDE.
• **Fluid Animations:** Code-driven React Native Skia & Reanimated 3 engines. Ensures zero frame-drop liquid physics without large video assets.
• **Backend & Storage:** Supabase PostgreSQL with Row Level Security (RLS). Images uploaded to private Supabase buckets prior to AI routing.
• **AI Processing:** OpenRouter during active development/testing; seamless cutover to Gemini Flash in production for cost-efficient vision parsing.

7. Go-To-Market & Organic UGC Marketing Strategy

Rather than burning early capital on saturated digital ad auctions, Nutro's user acquisition relies on viral short-form demonstration loops across **Instagram Reels and YouTube Shorts**.

CONTENT PILLAR A: "FRIDGE TO FEAST"

Hook: *"POV: It's 10 PM, your fridge is almost empty, but this AI just made a 400-calorie dinner."*
Execution: Real-life smartphone recording of messy fridge $\rightarrow$ Nutro scan screen $\rightarrow$ Gemini outputs instant paneer-bhurji/stir-fry recipe matching remaining macros.
Audience: Busy young professionals, college students.

CONTENT PILLAR B: "MACRO REALITY CHECK"

Hook: *"Everyone told me Indian ghar ka khana has zero protein. Let's see what happens when I scan my thali."*
Execution: Scanning authentic local plates (dal, roti, sabzi, curd) to demonstrate instant, accurate regional food breakdown.
Audience: Fitness enthusiasts across Indian metro areas.

CONTENT PILLAR C: "AESTHETIC FLUID HABITS"

Hook: *"Tracking water used to be a chore until my phone did this with every sip."*
Execution: Close-up macro video of high-refresh rate Skia liquid animations reacting to dynamic drinking intervals.
Audience: Habit trackers, aesthetic lifestyle accounts.

CONTENT PILLAR D: "FAMILY SYNC"

Hook: *"How I monitor my kids' water intake during hot exam months without asking them 10 times."*
Execution: Demonstrating Nutro Home's multi-profile orbit dashboard syncing live between parent and child devices.
Audience: Parents, health-conscious family admins.

8. Pre-Launch Validation: Waitlist Funnel on nutro-ai.app

Phase	Deliverable	Primary Metric / KPI
Phase 2.5 Landing Page	Next.js 15 + Tailwind single-page waitlist hosted on Hostinger / Vercel pointing to `nutro-ai.app`.	> 25% visitor-to-waitlist email conversion rate.
Pre-Launch UGC Pulse	5 organic short-form reels showcasing the fridge AI & liquid cup micro-demo.	1,000+ waitlist registrations before publishing to TestFlight.
Beta Deployment	Expo EAS internal distribution for iOS TestFlight & Android Play Console Closed Testing.	Crash-free session rate > 99.2%; AI accuracy rating > 90%.